



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER



Test Report No. 2023 – 0001



Agricultural Machinery – Rotary Sifter JHT RS-2

Test Report Number	2023 – 0001
Date of Issuance	March 28, 2023
Valid Until	March 28, 2028
Agricultural Machine	Rotary Sifter
Type	Rotary, Drum type, Continuous type, For vermicast
Brand	JHT
Model	RS-2
Test Requested by	JHT MICRO ENTERPRISES Brgy. San Pascual, Talavera, Nueva Ecija

SUMMARY

Performance Criteria	Manufacturer's Specification	AMTEC Test
Input Capacity, kg/h	ND ^a	2278.77
Sifting Capacity, kg/h	ND ^a	2209.36
Vermicast Recovery, %	ND ^a	89.47

^a Not provided by the requesting party

Note: ND – No Data

OBSERVATIONS

1. Operator's manual (based on PAES 102:2000)	
a. Language	English
b. Safety and warnings	Provided
c. Operating information	Provided
d. Accessories and attachments	Not provided
e. Maintenance instruction	Provided
f. Storage	Provided
g. Handling, reception, transportation, assembly, and installation	Provided
h. Specifications	Not provided
i. Dismantling and disposal	Not provided
j. Warranty	Provided
k. Parts list	Provided
2. Previous test reports of similar brand and model	None
3. Manufacturer	JHT MICRO ENTERPRISES
4. Prime mover used	4.1 kW HONDA GP160 Gasoline Engine
5. Number of operators	Three (One feeder, two baggers)
6. Loading of input	The input was placed on the input chute which was accessible to the operator.
7. Cleaning of parts	All parts are accessible for cleaning.
8. Adjusting and repairing of parts	Locally manufactured parts are repairable.
9. Collecting of output	The output fell on a tarpaulin mat which was later collected by the baggers.
10. Transporting the machine	The machine is mobile with two cart wheels at the front.
11. Safety features	The transmission system is provided with cover.
12. Failures or abnormalities that may be observed on the machine or its component parts during and after the milling operation	None
13. Other remarks	None

PURPOSE AND SCOPE OF TEST

Purpose	Performance Testing / Commercial
Date of Test	March 15, 2023
Location of Test	Brgy. San Pascual, Talavera, Nueva Ecija 15°36'55.96" N, 120°54'47.26" E
Items Tested	Operator's Manual, Input Capacity, Sifting Capacity, Vermicast Recovery, Fuel Consumption, Speed of Components, Noise Level, and Number of Operators

METHODS OF TEST

The rotary sifter was tested following the AMTEC Methods of Test.

DESCRIPTION OF THE MACHINE

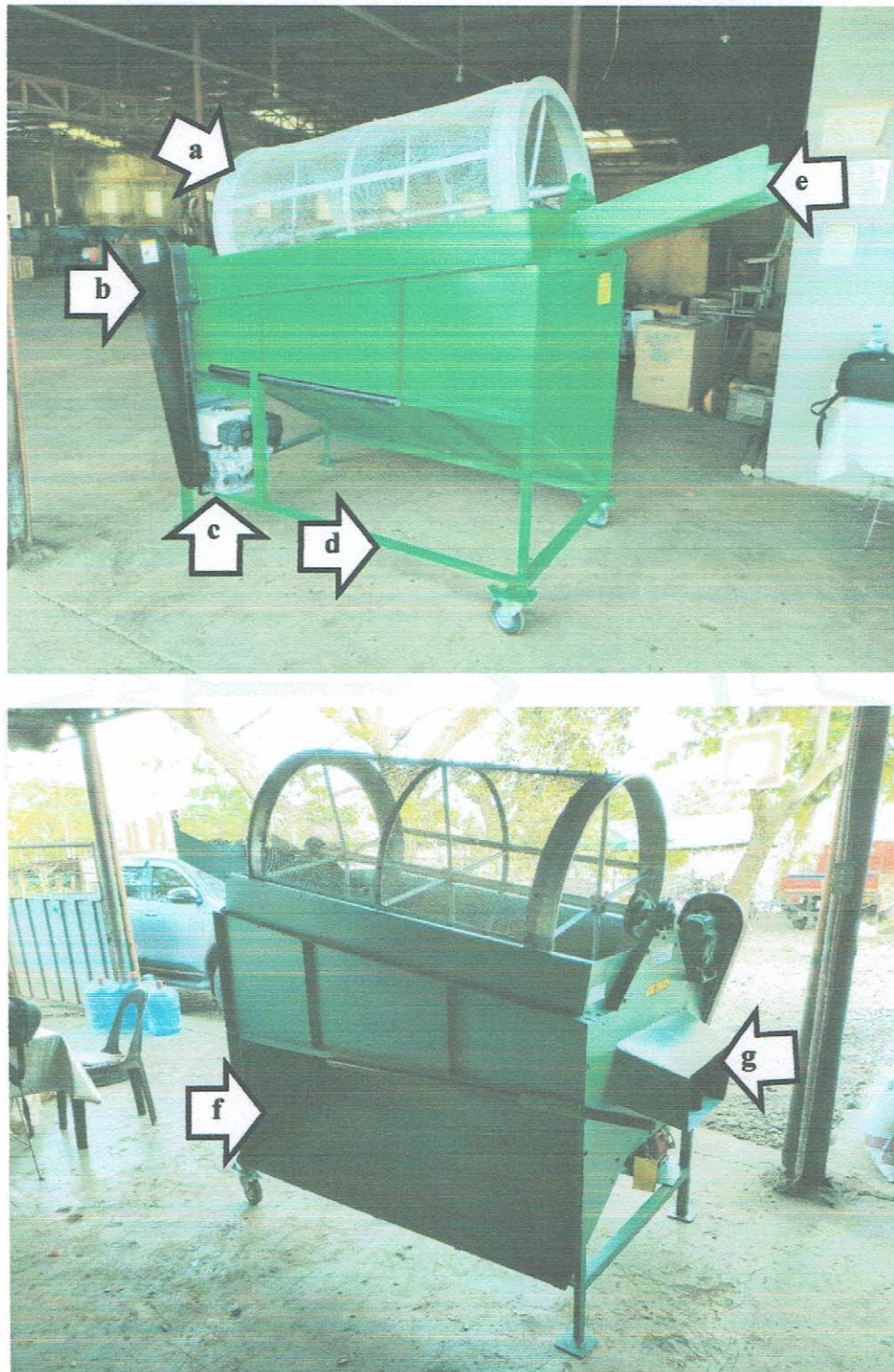


Figure 1. Parts of the JHT RS-2 Rotary Sifter powered by 4.1 kW HONDA GP160 Gasoline Engine include the following: (a) Rotary sieve, (b) Belt and pulley transmission system with cover, (c) Prime mover, (d) Main frame, (e) Input chute, (f) Vermicast output chute, and (g) Worm output chute.

SPECIFICATIONS

Item		Manufacturer's Specification ^a	AMTEC Verification
1	Overall dimensions and weight		
1.1	Length, mm	ND	2405
1.2	Width, mm	ND	1128
1.3	Height, mm	ND	1920
1.4	Weight, kg	ND	NM ^b
2	Hopper		
2.1	Dimension, L × W × H, mm	ND	753 × 530 × 132
2.2	Height from the ground, mm	ND	1480
2.3	Angle of friction, °	ND	9
2.4	Material	ND	Unknown metal ^c (t = 1.36 mm)
3	Rotary sieve		
3.1	Dimension, L × D, mm	ND	1367 × 780
3.2	Material	ND	Unknown metal ^c
4	Output chute		
4.1	Dimension, L × W × H, mm	ND	555 × 600 × 186
4.2	Dimension of opening, L × W, mm	ND	210 × 345 × 186
4.3	Height from ground, mm	ND	627
4.4	Material	ND	Unknown metal ^c (t = 1.36 mm)
5	Handle		
5.1	Dimensions, L × D, mm	ND	1220 × 33.46
5.2	Material	ND	Unknown metal ^c
6	Main frame		
6.1	Dimensions, L × W × H, mm	ND	1630 × 990 × 1260 (Angle bar, t = 2.69)
6.2	Material	ND	Iron/Carbon steel ^c (t = 2.69 mm)

^a The requesting party did not submit a complete manufacturer's specification.

^b The data were not measured due to the unavailability of instrument that will capacitate the measurement needed.

^c Determined using the BRUKER S1 TITAN 800 X-ray Fluorescence Analyzer. See Tables 1 to 5 for elemental composition.

Note: ND – No Data, NM – Not Measured

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specification ^a	AMTEC Verification
7	Engine		
7.1	Brand	HONDA ^b	HONDA ^b
7.2	Model	GP160 ^b	GP160 ^b
7.3	Make	ND	ND
7.4	Rated power, kW	4.1 ^b	4.1 ^b
7.5	Rated speed, rpm	3600 ^b	3600 ^b
7.6	Weight, kg	ND	NM ^c
8	Power transmission system		
8.1	Engine to speed reducer drive gear		
8.1.1	Engine ^d	ND	50.8 × 1B × 19.32
8.1.2	Speed reducer drive gear ^d	ND	203.2 × 1B × 19.74
8.1.3	Belt size	ND	B-83
9	Safety devices	ND	None
10	Special features	ND	None

^a The requesting party did not submit a complete manufacturer's specification

^b The data were obtained from the engine's nameplate.

^c The data were not measured due to the unavailability of instrument that will capacitate the measurement needed.

^d Pulley diameter, mm × number of belts × shaft diameter, mm

Note: ND – No Data, NM – Not Measured

RESULTS

Item		Data	
1	Condition of test materials		
1.1	Test material	Vermicast with African Night Crawler Worms	
1.2	Source	Not provided	
1.3	Vermicast Moisture content, % _{wb}	36.30	
1.4	Vermicast concentration, %	88.97	
1.5	Dimensions of worms, mm		
1.5.1	Length,	115.6	
1.5.2	Diameter	9.80	
2	Performance test		
2.1	Weight of input, kg	245.9	
2.2	Input capacity, kg/h	2278.77	
2.3	Weight of output, kg	220.0	
2.4	Output capacity, kg/h	2043.43	
2.5	Vermicast recovery, %	89.47	
2.6	Sifting Capacity, kg/h	2209.36	
2.7	Speed of rotating components, rpm	Without load	With load
2.7.1	Engine shaft	1575	1572
2.7.2	Speed reducer drive gear	431	429
2.7.3	Speed reducer driven gear	7	7
2.7.4	Rotary sieve shaft	14	14
2.8	Noise level at ear level, dB(A)	Without load	With load
2.8.1	Feeding operator	74.6	72.0
2.8.2	Soil output collector	76.1	72.6
2.8.3	Worm output collector	78.3	77.5
2.9	Fuel consumption rate, L/h	0.215	
2.10	Vermicast concentration from output, %		
2.10.1	Output from worm outlet	79.04	
2.10.2	Output from vermicast outlet	100.00	

Note: NA – Not Applicable

RESULTS (Cont'n)

Table 1. Elemental composition of the hopper of JHT RS-2 Rotary Sifter.

Element	Percent Composition^a
Aluminum	<LOD
Silicon	<LOD
Phosphorus	1.22
Titanium	<LOD
Vanadium	<LOD
Chromium	0.13
Manganese	0.15
Iron	85.20
Cobalt	0.38
Nickel	<LOD
Copper	0.03
Niobium	0.01
Molybdenum	0.08
Tin	0.05
Tungsten	12.61
Lead	<LOD
Arsenic	<LOD
Zirconium	0.05

^a Determined using BRUKER S1 TITAN 800 X-ray Fluorescence Analyzer.

Note: LOD – Limit of Detection

Table 2. Elemental composition of the output chute of JHT RS-2 Rotary Sifter.

Element	Percent Composition^a
Aluminum	0.38
Silicon	0.47
Manganese	<LOD
Iron	13.05
Nickel	<LOD
Copper	<LOD
Tin	0.03
Lead	0.04
Zinc	86.00
Bismuth	0.03

^a Determined using BRUKER S1 TITAN 800 X-ray Fluorescence Analyzer.

Note: LOD – Limit of Detection

RESULTS (Cont'n)

Table 3. Elemental composition of the handle of JHT RS-2 Rotary Sifter.

Element	Percent Composition ^a
Aluminum	<LOD
Silicon	0.07
Manganese	0.20
Iron	49.78
Nickel	0.01
Copper	<LOD
Tin	<LOD
Lead	0.38
Zinc	49.19
Bismuth	0.33

^a Determined using BRUKER S1 TITAN 800 X-ray Fluorescence Analyzer.

Note: LOD – Limit of Detection

Table 4. Elemental composition of the main frame of JHT RS-2 Rotary Sifter.

Element	Percent Composition ^a
Aluminum	0.00
Silicon	0.47
Phosphorus	<LOD
Sulfur	<LOD
Titanium	0.01
Vanadium	0.05
Chromium	0.03
Manganese	0.24
Iron	98.90
Cobalt	<LOD
Nickel	0.05
Copper	0.06
Niobium	<LOD
Molybdenum	0.05
Tin	0.08
Tungsten	<LOD

^a Determined using BRUKER S1 TITAN 800 X-ray Fluorescence Analyzer.

Note: LOD – Limit of Detection

OBSERVATIONS



Figure 2. Markings and labeling of the JHT RS-2 Rotary Sifter.



Figure 3. Quality from worm output chute

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